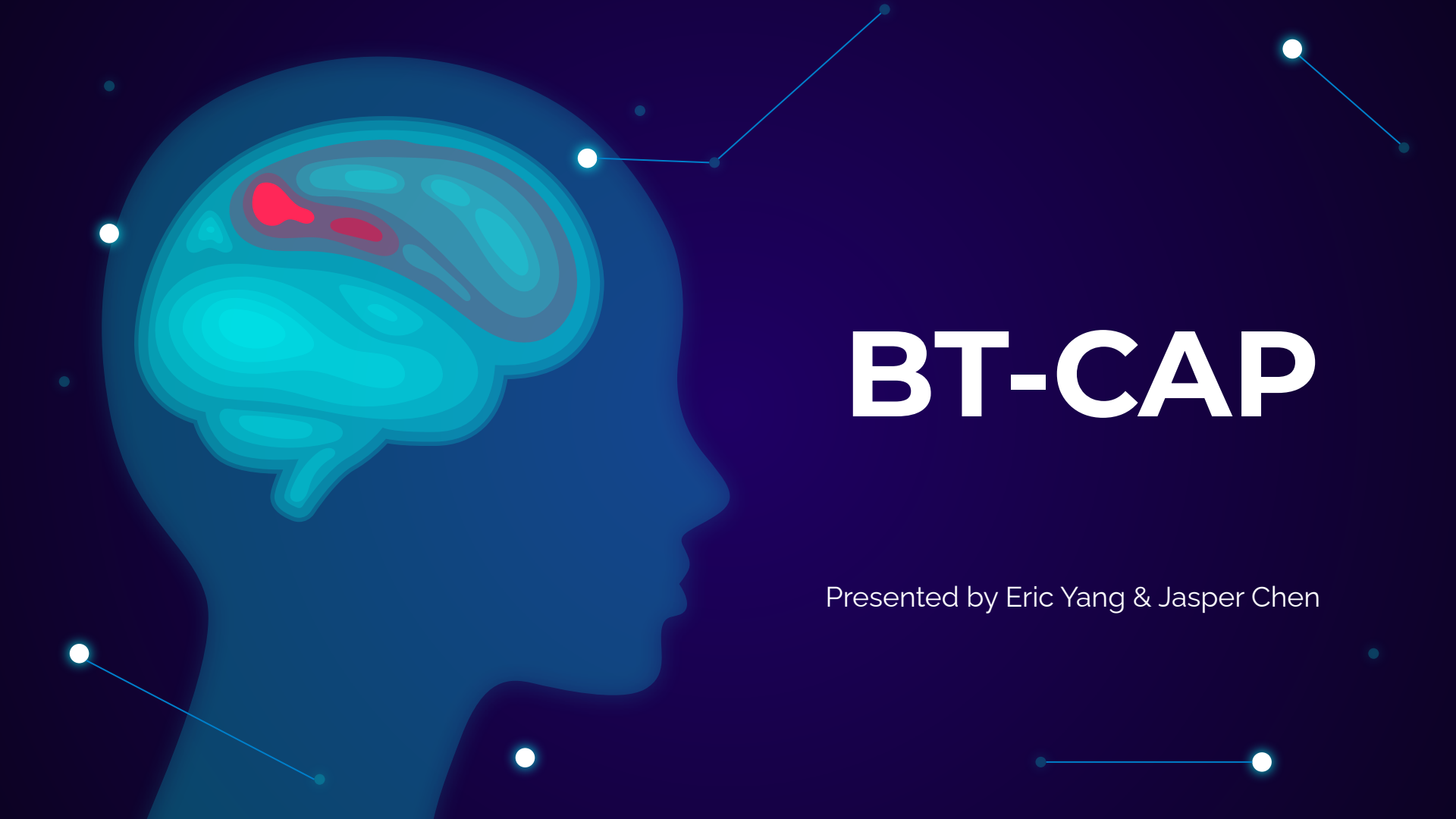


BT-CAP

Presented by Eric Yang & Jasper Chen

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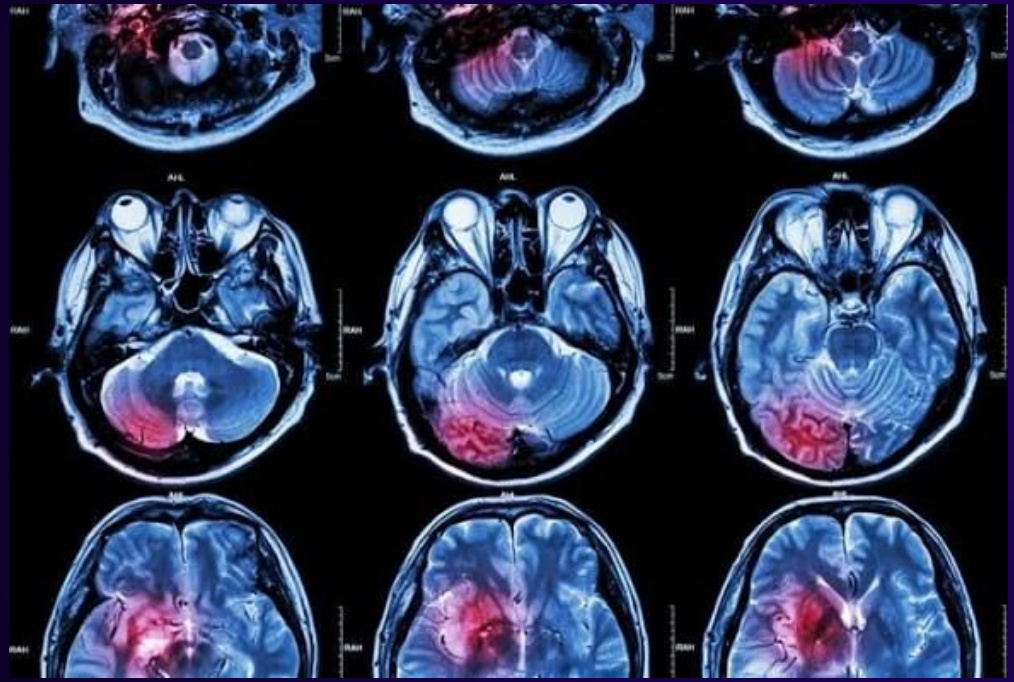
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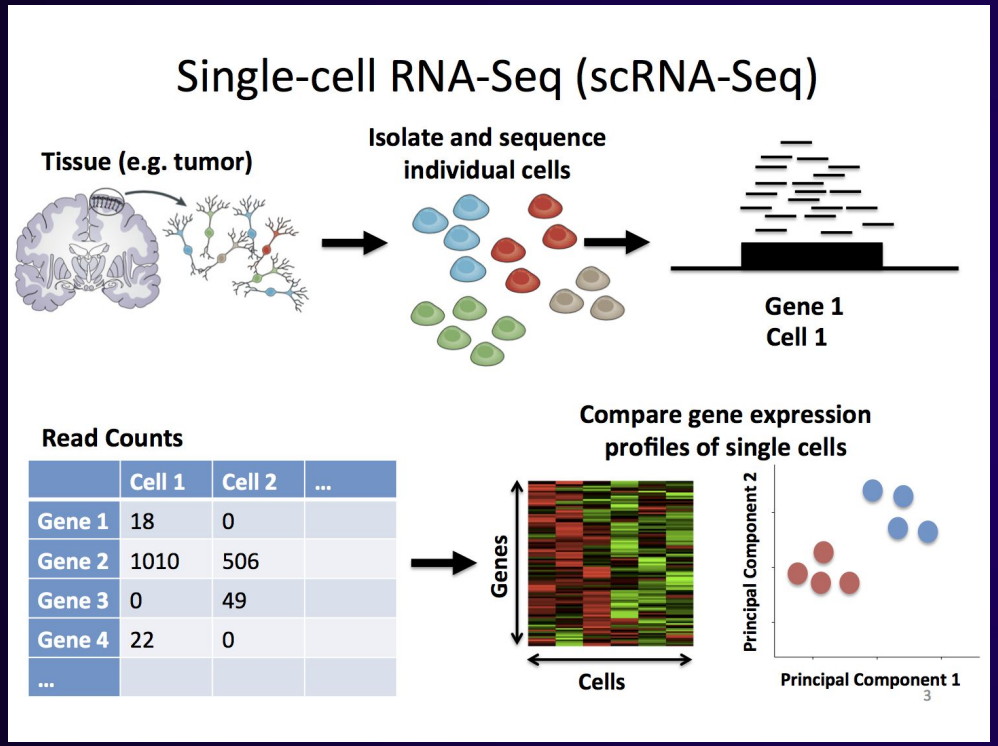
Introduction





Brain Tumors

An abnormal mass of cells that disrupts the brain's delicate ecosystem. For our platform, we focus on Primary Brain Tumors that originate within the Central Nervous System (CNS).



	Cell 1	Cell 2	...
Gene 1	18	0	
Gene 2	1010	506	
Gene 3	0	49	
Gene 4	22	0	
...			

Single-Cell RNA Sequencing

Genomic technology that measures the gene expression (transcriptome) of thousands of individual cells simultaneously. Unlike "bulk" sequencing, which averages the signal of an entire tissue, scRNA-seq provides a high-resolution map of every distinct cell type present.

Biological Complexity & Clinical Stakes

Challenge of Heterogeneity

Brain tumors (gliomas, medulloblastomas) are "mosaics" of malignant, immune, and stromal cells that vary significantly within a single patient.

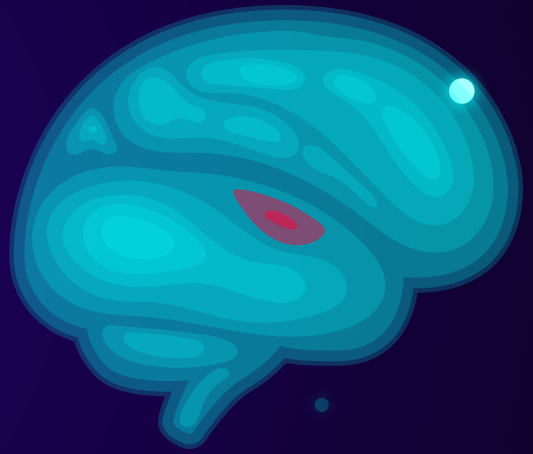
Cellular Plasticity

Malignant cells constantly shift between states (e.g., stem-like to progenitor), allowing them to evade standard therapies.

Discovery Bottleneck

Identifying rare, aggressive cell populations is the "first mile" of precision medicine; if we can't label them, we can't target them.

How can we leverage transformer-based foundation models and high-dimensional vector search to overcome the limitations of manual annotation and accurately map the complex cellular 'mosaic' of the brain tumor microenvironment?



2

Current Solutions



Limitations of Existing Approaches

Standard LLM Hallucinations

Out-of-the-box LLMs struggle with raw genomic context.

MedGemma 1.5-4B truncated output on ~40% of queries and hallucinated biological relationships.

Retrieval Flaws

Standard TF-IDF retrieval buries exclusive cellular pathways under redundant hits.

Heuristic Limitations

Single-table rules (DEG log2FC or raw CellChat edges) over-index on raw expression rather than functional context.

Single-column baselines hit an AUROC ceiling of 0.51–0.58.

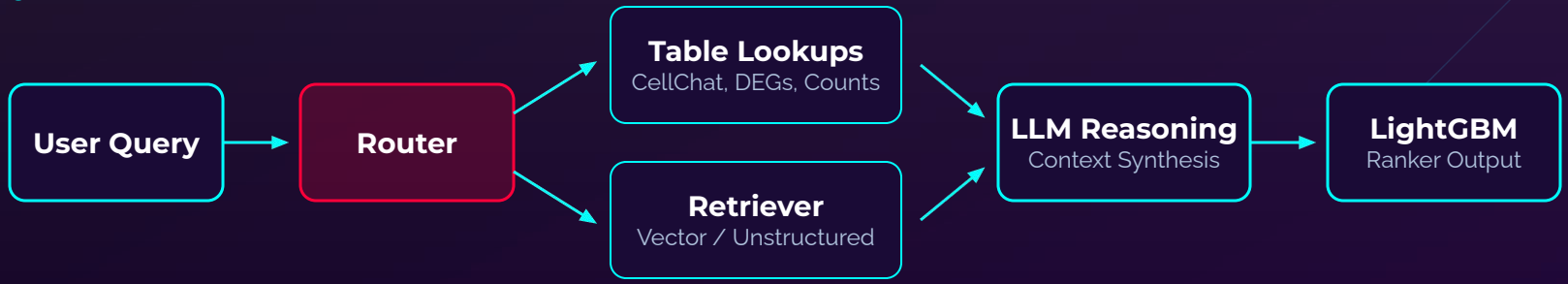
3

Proposed Solution



Agentic Bio-Reasoning Framework

System Architecture Flow



Key Takeaway: Deterministic table-lookup tools (CellChat, DEGs, cell counts) anchor the agent directly to structured in-house data.
By constraining the reasoning pipeline with verified tabular metrics before synthesizing context through the LLM, the framework effectively **eliminates biological hallucinations** and ensures mathematically grounded outputs.

Dual-LLM Reasoning with Gemini 2.5 Flash

MedGemma Evaluation

- Required dedicated Vertex endpoints (incurred API cost).
- Truncated responses and loops on complex multi-step questions.

Gemini 2.5 Flash Selection

- Clean, structured outputs separating observation from biological interpretation.
- Accurate, cited answers across all 24 test questions without inventing pairs or DEGs.

Capturing Context with scETM Gene Embeddings

Scaling Universe

Expanded gene annotation set dramatically to broaden analytical scale:

33 → 9,580

Genes labeled via SynGO and QuickGO (GO:0001772) for robust functional coverage.

32-D Vectors

High-dimensional representations extracted from massive single-cell profiles:

20,000+

Single cells embedded into 32-dimensional scETM vector space to capture latent co-expression networks and functional states.

4

Performance Metrics



Limitations of Existing Approaches

- Tumor Synaptic (Tables + scETM): 0.742 AUROC | 0.437 AUPRC
- Immune Synapse (Tables + scETM): 0.725 AUROC | 0.021 AUPRC
- 3-Table Rule Baseline: 0.631 AUROC
- Raw DEG log2FC Baseline: 0.612 AUROC
- Takeaway: Adding scETM co-expression embeddings boosted AUROC by +0.06 to +0.07 over table-only features.

Model	Arm	AUROC	AUPRC	Housekeeping genes in top 100	Published-33 median percentile
Tumor synaptic	Tables only	0.675	0.332	83	0.98
Tumor synaptic	Tables + scETM	**0.742**	**0.437**	83	0.82
Tumor synaptic	scETM only	0.738	0.433	83	0.72
Immune synapse	Tables only	0.666	0.015	0	0.91
Immune synapse	Tables + scETM	**0.725**	**0.021**	0	0.57
Immune synapse	scETM only	0.706	0.020	0	0.49

Performance: Surfacing True Biological Signal Over Noise

Before (Tables Only)

- Top ranks dominated by off-list mitochondrial and ferritin genes.
- High technical background noise obscures specific driver signals.

→

After (With scETM Embeddings)

- Filtered metabolic noise effectively across datasets.
- **78/83** top housekeeping hits correctly identified as ribosomal genes with SynGO labels.

Tumor Synaptic Nominees

GRIA2, NRXN1, PTPRZ1; identified novel targets NCL, GRIK2, BSG, MPZL1, CADM2.

Immune Synapse Nominees

CD28, HAVCR2, LCP2, GRAP2, and MHC-II markers (HLA-DRA, CD74).

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5

Future Steps



Final Weeks Roadmap & Hand-Off Goals

MIDTERM
Oct 2026

PORTAL INTEGRATION
Nov 2026

FINAL HAND-OFF
Dec 4, 2026

01

Portal Chatbot Integration

Deploy the bio-reasoning agent as an interactive chatbot component in the BT-CAP web portal (code on GitHub).

02

LLM Consistency Testing

Integrate 5x repeated local Ollama/MedGemma runs into --llm-runs to validate output stability.

03

Final Refinements

Apply team-reviewed literature filters, conduct a myeloid contamination check, and finalize hand-off documentation by December 4th.